Statistical Texture Awareness Network for Hyperspectral Image Classification

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***Abstract*— The distribution of ground objects in hyperspectral images predominantly reveals spatial indications of both order and disorder, encapsulating a wealth of texture information. This texture information encompasses not only local structural details but also global statistical priors of an image. Nevertheless, convolutional-neural-network-based methods for hyperspectral image classification (HIC) primarily use skip connections to incorporate shallow features abundant in texture information into deeper layers. They face challenges in effectively capturing the statistical properties of texture information, and the traditional method of modeling statistical attributes struggles to seamlessly integrate into parameter learning of convolutional neural networks (CNNs). To do so, this work proposes a statistical texture awareness network (STANet) for HIC. It achieves the exploration of learnable texture features. Through multilevel quantization and quantization encoding, a statistical texture learning module (STLM) is constructed to represent texture information from low-level features in a statistical manner. As a result, it augments the discriminatory power of such features. In addition, a complete feature fusion module (CFFM) is designed to intelligently combine multiscale contextual semantic and statistical texture features, thereby bolstering the discrimination of spectral–spatial ones. Experimental results reported for three public datasets demonstrate the superior performance of the proposed network over other peers.**

***Index Terms*— Convolution neural network, feature extraction, hyperspectral image classification (HIC), semantic feature, statistic feature.**

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I. Introduction

REMOTE sensing land cover classification is the process of analyzing remote sensing images to categorize pixels or regions into different land cover categories. It is widely used in practical applications such as precision agriculture [1], environmental monitoring [2], and mineral exploration [3]. Hyperspectral remote sensing images encompass fine-grained spectral information and rich spatial details, which can make the classification of land covers more precise. This has promoted hyperspectral data analysis to become active [4], [5], [6], [7], [8], [9], [10], [11], [12], especially hyperspectral image classification (HIC) [13], [14], [15], which is the basis of remote sensing land cover classification. Unlike red–green–blue (RGB) images, hyperspectral images contain a large number of spectral bands, have high data dimensions, and are susceptible to atmospheric radiation. Specifically, the inherent challenges of spectral similarity among different objects and spectral differences among similar objects pose difficulties in HIC. Relevant scholars in the hyperspectral community are committed to alleviating the above problems from the perspective of feature extraction, which has become the focus of research in HIC.

During the initial phases of hyperspectral image feature extraction development, spectral features preprocessed through principal component analysis [16] or linear discriminant analysis [17] are used as an input for traditional classifiers [18], [19], [20], [21] to predict land cover labels. Notably, the predominant focus on spectral features might overlook the vital relationship between neighboring pixels in a spatial domain. As a remedy, the spectral–spatial feature extraction approach has emerged as a prevailing trend, encompassing the synergistic integration of both the spectral and spatial attributes. Successive advancements in HIC build on spectral–spatial features by incorporating a variety of techniques such as handcrafted features' modeling [22], [23], [24], [25], tensor analysis [26], and transform domain methods [27], [28], [29] are harnessed to effectively characterize the spectral–spatial characteristics of hyperspectral images. Nonetheless, these methods depend on the particular strategies devised by the designer, which rely heavily on their prior expertise. Regrettably, such methods encounter challenges in adequately capturing the intrinsic attributes of objects and lack robustness when dealing with high-dimensional and redundant hyperspectral data.

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The advent of deep learning technology has ushered in a new epoch for the spectral–spatial feature extraction of hyperspectral images [30], [31], [32]. As a seminal milestone in the domain of deep learning, the residual network [33] significantly propels the advancement of CNNs in the extraction of spectral–spatial features from hyperspectral images [34], [35]. A multidirection network is proposed to capture more profuse array of spectral–spatial information through multidirection sampling [36]. A spectral–spatial residual network (SSRN) is designed, using 3-D convolution to holistically model spectral–spatial features and alleviate the shortcomings of 2-D convolution that fails to fully capture spectral features [37]. Roy et al. [38] come up with hybrid spectral CNN (HybridSN) by amalgamating 2-D and 3-D convolution. To address the challenge of limited training samples inadequately training models with numerous parameters, LiteDepthwiseNet [39], GhostNet [40], and Gabor ensemble filter [41] are introduced to diminish the model's parameter count. Introducing attention mechanisms [42], [43], [44], [45] to extract more discerning spectral–spatial features has become pervasive. A residual spectral–spatial network (RSSAN) refines spectral–spatial features adaptively, leveraging spectral and spatial attention (SA) mechanisms [46]. An attention-based adaptive spectral–spatial kernel ResNet (A2S2KNet) models cross-channel dependencies using attention mechanisms while effectively tuning neuron receptive fields [47]. Li et al. [48] devise a center vector oriented self-similarity network (CVSSN) to mine the relationship between a center pixel and its neighborhood in patch. Li et al. [49] propose the hierarchical homogeneity-attention network to rectify the lack of supervision on the target region location inherent in self-attention. Furthermore, graph convolutional network [50], generative adversarial network [51], recurrent neural network [52], and Transformer [53] have found extensive applications in representing spectral–spatial information.

Several HIC methods based on CNN paradigms and grounded in awareness strategies are proposed, thereby fostering advancements in HIC. A group-aware hierarchical Transformer is proposed to emphasize local relationships within spectral channels by introducing a grouped pixel embedding module [54]. Context information awareness in hyperspectral images is a popular strategy. An adaptive superpixel segmentation technology is proposed to mine local and global context information, achieving awareness of multiscale context information in hyperspectral images [55]. Tu et al. [56] propose a fully convolutional network (FCN) for nonlocal-dependent learning to enhance awareness of context features in the spectral domain. To alleviate the disturbance of adversarial samples on the model, a robust context-aware network is proposed. It learns the global context representation through aggregating dilated convolution and constructs a class context learning module to explicitly model intraclass and interclass context information [57].

Capturing contextual information in non-Euclidean space is another approach. Autoregressive moving average filters and context-aware learning are used for HIC, which can better capture the global graph structure and have stronger robustness to noise [58]. Self-correlated learning smoothy enhanced locality-preserving graph convolution embedding clustering is constructed to enhance the robustness and adaptability of clustering methods by being aware of smoothy information and nonlocal relationships in the hidden representation space, improving the feature expression ability [59]. Ding et al. [60] introduce a spectral–spatial transformation preprocessing mechanism and propose a locally preserved low-pass graph convolutional embedding autoencoder to perceive smoother features in hyperspectral images and maintain layered local features. Normalized spectral clustering with kernel-based learning is proposed to achieve distinguishable embedding by aggregating local-to-global correlations, effectively perceiving spatial contextual features in non-Euclidean space [61]. Jiang et al. [62] propose a graph generation structure-aware Transformer to obtain the absolute position sequence of pixels, introducing explicit physical meaning to position encoding. Hyperspectral images contain rich spatial information, and efficient perception of spatial texture information is crucial to improving classification accuracy. Zhang et al. [63] propose a feature extraction method for structural and textural awareness to enhance the perception of the intrinsic structure and texture of hyperspectral images. Li et al. [64] construct a texture-aware self-attention model that improves spatial contrast and overcomes spectral variance through feature pyramids and Gabor features, achieving accurate classification of tree species.

A significant portion of the aforementioned methodologies uses image patch sampling, that is, selecting local neighborhood image patches centered at the pixels to be classified. As depicted in Fig. 1(a), the patch-based sampling approach faces challenges in adeptly capturing long-range dependencies. The existence of overlapping sampling results in redundant computations. Furthermore, the selection of patch size directly impacts the classification performance. HIC bears resemblance to the scene parsing of natural images, aiming to segment different categories of regions within the image [65], [66], [67], [68], [69]. Differently, HIC aims to assign a specific label to each pixel, which belongs to pixelwise image classification. By drawing inspiration from the scene parsing of natural images, an FCN [70] is introduced into HIC. It liberates the model from being constrained by the selection of neighbor image patch size, fostering new opportunities for performance enhancement. Xu et al. [71] propose an end-to-end spectral–spatial fully convolution network (SSFCN), which eschews repetitive patch sampling and exhibits superior efficiency compared with patch-based methodologies. Zheng et al. [72] introduce a fast patch-free global learning approach called FreeNet, comprising a spectral attention encoder and a lightweight decoder. Jiang et al. [73] combine 3-D fully convolution and spatial propagation network to present a fully convolutional spatial propagation network. Shen et al. [74] incorporate an efficient nonlocal module into FCN (ENL-FCN) to extract long-range context information between pixels. Furthermore, Wang et al. [75] extend whole image-based HIC to hyperspectral scene parsing, proposing the fully contextual network (FContNet) with two multiscale

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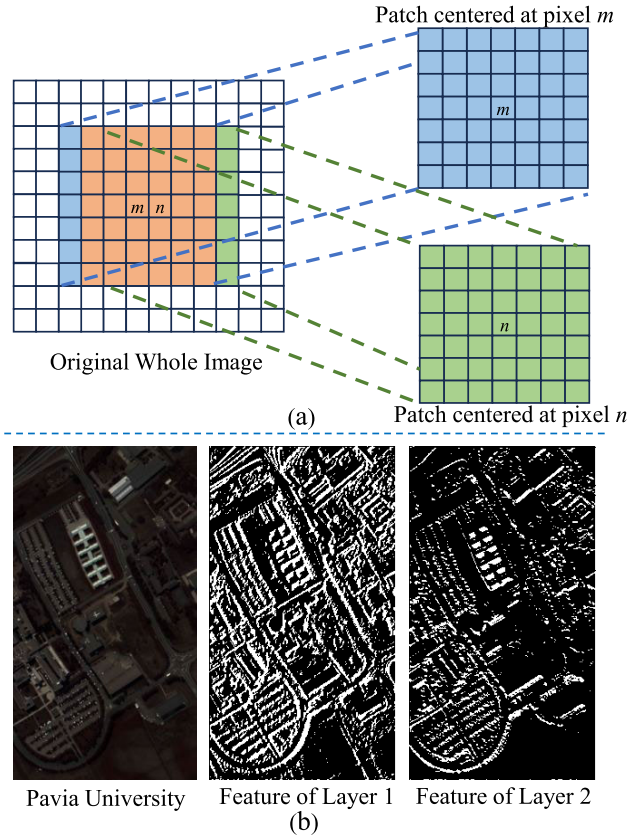


Fig. 1. Schematic of patch-based sampling and shallow features. (a) Neighborhood patches of two adjacent centers. (b) Shallow feature maps of the Pavia University dataset.

feature extraction modules, namely, FContNet-Pyramid and FContNet-ASPP.

However, the aforementioned approaches for HIC center around FCN to investigate contextual information within high-level features. They exclusively leverage high-level features derived from an extensive receptive field, overlooking pivotal low-level intricacies such as edges. As illustrated in Fig. 1(b), the distribution of land covers in hyperspectral remote sensing images manifests characteristics of both order and disorder, with shallow features encapsulating a wealth of texture information. The texture information encompasses not only local structural properties, such as edges but also global statistical properties such as the distribution of pixel values or regional characteristics. The statistical attributes of texture information play a significant role in image analysis [76], [77]. Certain methodologies incorporate statistical texture features into HIC tasks, yet these approaches embrace an explicit, handcrafted feature modeling paradigm [22], [23], [29]. Nonetheless, within the CNN framework, there exists no mechanism to explicitly extract the statistical texture information inherent in hyperspectral images. Handcrafted feature construction relies on preset parameters or templates, posing challenges in effectively integrating them into parameter learning of the CNN framework.

To mitigate the aforementioned challenges, this work proposes a statistical texture awareness network (STANet), which is designed to represent and explore learnable statistical texture features in hyperspectral images. As a result, comprehensive semantic-texture features are extracted. In STANet, the statistical texture learning module (STLM) is proposed to encode texture statistical features by performing multilevel quantization statistics on low-level features rich in texture information. Subsequently, multiscale semantic features are mined through an atrous spatial pyramid pooling (ASPP) module. Finally, a complete feature fusion module (CFFM) is designed to adaptively fuse the two sets of features to improve discrimination. Therefore, the main contributions of this article are summarized as follows.

1) It proposes STANet that effectively integrates statistical texture feature and high-level semantic features through designed CFFM module. STANet enhances the completeness of captured features by concurrently extracting both statistical texture features and semantic features in an end-to-end fashion.
2) It establishes STLM that effectively learns the texture statistical properties of the low-level features through multilevel quantization statistics. This module seamlessly incorporates traditional statistical methods into CNN architecture.

The originality of this article resides in bridging the divide between the statistical attributes of manually crafted modeling of texture information and the learnable CNN framework in HIC tasks. The acquisition of statistical texture properties from shallow features is seamlessly integrated into the CNN framework. This network can fully explore statistical texture and semantic information of hyperspectral images at the feature level.

Section II introduces the proposed network. Experimental results and analyses are presented in Section III. Finally, the conclusion for this article is given in Section IV.

II. Proposed Methodology

In this section, the proposed STANet is presented in detail. Fig. 2 illustrates STANet's overall framework. A hyperspectral image is fed into STANet in its entirety without neighborhood cropping. First, a spectral–spatial feature extraction module, which includes SA and channel attention (CA), is designed to extract spectral–spatial features from hyperspectral images. To extract statistical texture information, STLM is devised to perform multilevel quantitative statistical representation of low-level features. ASPP is used to explore the multiscale semantic features of hyperspectral images. Finally, we introduce an adaptive feature fusion module named CFFM to adaptively combine statistical texture features and semantic features.

A. Statistical Texture Awareness Network

As shown in Fig. 2, STANet directly inputs the whole image into the trained model and generates the classification results. It consists of three parts: semantic feature extraction, statistical texture feature extraction, and feature fusion. In the process of semantic feature extraction, an input hyperspectral image is initially encoded by a 3×3 convolutional layer. Both CA and SA layers [78] with a residual connection structure are

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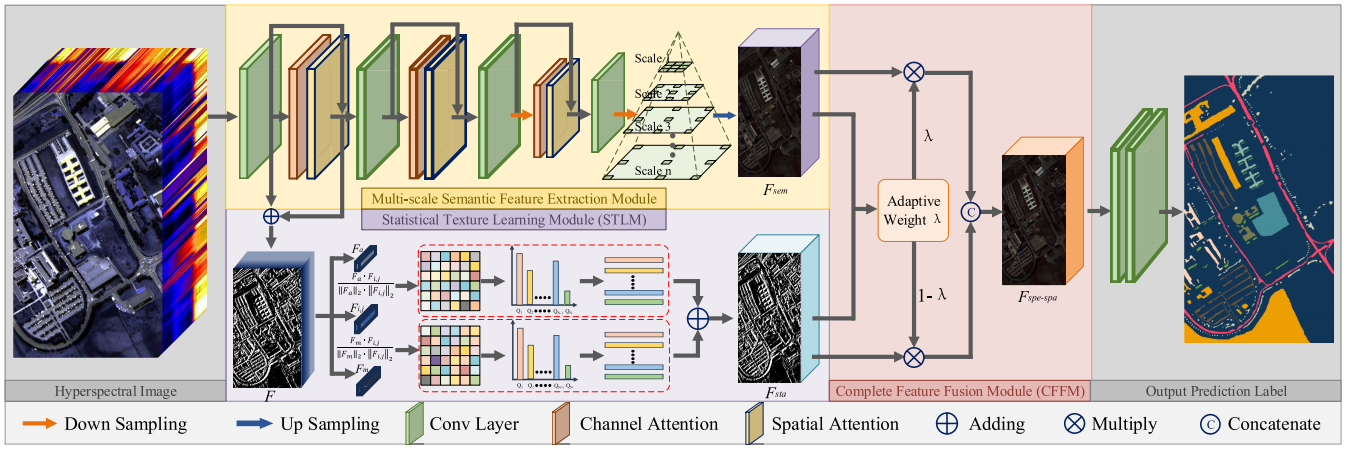


Fig. 2. STANet's architecture. A hyperspectral image is fed into multiscale semantic feature extraction module to obtain spectral–spatial semantic features. The low-level features of first two layers are sent to the STLM to extract statistical texture features. Finally, the semantic features and statistical features are adaptively fused by CFFM for label prediction.

used to capture spectral–spatial features. The two layers are represented as

$$\mathrm{CA} = \mathrm{Sigmod}\left(\mathrm{MLP}\left(F_a^c\right) \oplus \mathrm{MLP}\left(F_m^c\right)\right) \otimes F \tag{1}$$

$$\mathrm{SA} = Sigmoid\left(\mathrm{Conv}\left(\mathcal{C}\left(F_a^s, F_m^s\right)\right)\right) \otimes F \tag{2}$$

where $F_a^c \in \mathbb{R}^{C\times 1\times 1}$ and $F_m^c \in \mathbb{R}^{C\times 1\times 1}$ represent the feature map $F \in \mathbb{R}^{C\times H\times W}$ after average pooling and max pooling in the spatial dimension, $F_a^s \in \mathbb{R}^{1\times H\times W}$ and $F_m^s \in \mathbb{R}^{1\times H\times W}$ denote the feature map after average pooling and max pooling in the channel dimension, C is the channel dimension, H and W are the space dimensions, MLP is a multilayer perceptron, Conv represents the convolutional layer, $\mathcal{C}$ denotes the concatenation operation, Sigmoid denotes the Sigmoid function, $\oplus$ is the elementwise adding operation, and $\otimes$ denotes the multiplication operation. Through attention and convolutional layers, average pooling downsampling is adopted to save computational resources. To capture semantic features of multiscale contexts, ASPP modules are embedded in the model. Due to the different sizes of the input hyperspectral image, an adaptive dilation rate d is set. The maximum dilation r is 75% of the input image size, and ASPP's dilation rate d is expressed as

$$d = [1, (r/2), (r/4), r] \tag{3}$$

$$r = \lfloor [\min(H, W)/2] * 0.75/2 \rfloor \tag{4}$$

where H and W are the hyperspectral image size, and $\lfloor\cdot\rfloor$ denotes a rounding down operation. Since the batch size is equal to 1, the group normalization (GN) is used. The activation function is rectified linear unit (ReLU). Before the semantic features are fed into CFFM, bilinear interpolation is used to restore the feature map size to its initial size.

In the process of semantic feature extraction, the features of the first two layers are fused and sent to STLM to capture texture statistical features of hyperspectral images. The semantic and statistical features are sent to CFFM to realize adaptive fusion of the two parts of features. The fused spectral–spatial features are classified to obtain the final classification map.

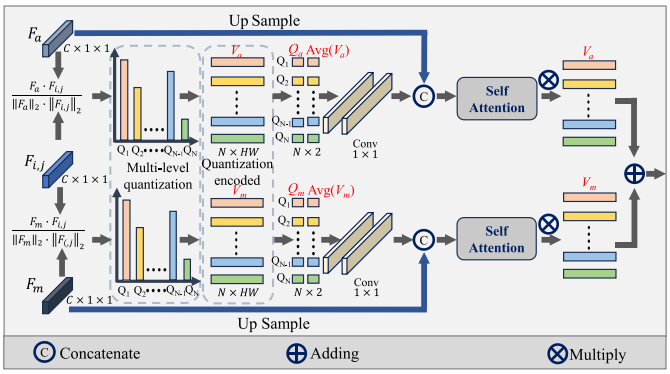


Fig. 3. STLM's architecture. For the low-level feature F, the similarity between F and the global average feature and the maximum feature are calculated, respectively. Then, the statistical texture features in the low-level features are mined by multilevel quantization and quantization coding to realize the learning of statistical texture features.

B. Statistical Texture Learning Module

Most whole image-based HIC methods are constructed using fully convolutional layers. While stacked convolutional layers excel in capturing local features with high-level semantic information, they fall short in adequately describing statistical textures. To address this limitation, STLM is proposed to effectively capture rich texture information from low-level features. STLM's flowchart is illustrated in Fig. 3.

The low-level feature map of STLM is denoted as $F \subset \mathbb{R}^{C\times H\times W}$. Its cosine similarity with global average feature $F_a \in \mathbb{R}^{C\times 1\times 1}$ and global maximum feature $F_m \in \mathbb{R}^{C\times 1\times 1}$, respectively, are defined as

$$S_{a_{i,j}} = \frac{F_a \cdot F_{i,j}}{\|F_a\|_2 \cdot \|F_{i,j}\|_2} \tag{5}$$

$$S_{m_{i,j}} = \frac{F_m \cdot F_{i,j}}{\|F_m\|_2 \cdot \|F_{i,j}\|_2} \tag{6}$$

where $F_{i,j}$ is the (i, j)th element located in F, and $S_a \in \mathbb{R}^{H\times W}$ and $S_m \in \mathbb{R}^{H\times W}$ are two cosine similarity matrices, respectively. Multilevel quantization is performed on the two similarity matrices. After S_a and S_m are reshaped to $\mathbb{R}^{HW}$, they

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are divided into N levels by multilevel quantization $Q_a \in \mathbb{R}^N$ and $Q_m \in \mathbb{R}^N$. Q_{a_n} and Q_{m_n} $(n \in [1, N])$ are formulated as

$$Q_{a_n} = \frac{S_{a_{\max}} - S_{a_{\min}}}{N} \cdot n + S_{a_{\min}} \tag{7}$$

$$Q_{m_n} = \frac{S_{m_{\max}} - S_{m_{\min}}}{N} \cdot n + S_{m_{\min}} \tag{8}$$

where $S_{a_{\max}}$ and $S_{a_{\min}}$ denote the maximum and minimum values of S_a, respectively. Similarly, $S_{m_{\max}}$ and $S_{m_{\min}}$ denote the maximum and minimum values of S_m, respectively. Subsequently, S_a and S_m are transformed into two quantization encoding matrices $V_a \in \mathbb{R}^{N \times HW}$ and $V_m \in \mathbb{R}^{N \times HW}$ that are formulated as

$$V_{a_{k,n}} = \begin{cases} 1 - |Q_{a_n} - S_{a_k}|, & -\frac{1}{2N} \le Q_{a_n} - S_{a_k} \le \frac{1}{2N} \\ 0, & \text{other else} \end{cases} \tag{9}$$

$$V_{m_{k,n}} = \begin{cases} 1 - |Q_{m_n} - S_{m_k}|, & -\frac{1}{2N} \le Q_{m_n} - S_{m_k} \le \frac{1}{2N} \\ 0, & \text{other else.} \end{cases} \tag{10}$$

In this way, the encodings of N quantization levels are concatenated to obtain V_{a_k} for $k \in [1, HW]$. In other words, the index of the maximum value of V_{a_k} and V_{m_k} reflects the quantization level of S_{a_k} or S_{m_k}. After obtaining V_a and V_m, each quantization level means a kind of texture information, statistical features $F_1 \in \mathbb{R}^{C \times N}$ and $F_2 \in \mathbb{R}^{C \times N}$ of texture information are captured by counting the quantization encoding, which can be represented as

$$F_1 = \mathcal{C}\left(\text{MLP}\left(\mathcal{C}\left(Q_a, \frac{\sum_{k=1}^{HW} V_{a_{k,n}}}{\sum_{n=1}^{N}\sum_{k=1}^{HW} V_{a_{k,n}}}\right)\right), F_a\right) \tag{11}$$

$$F_2 = \mathcal{C}\left(\text{MLP}\left(\mathcal{C}\left(Q_m, \frac{\sum_{k=1}^{HW} V_{m_{k,n}}}{\sum_{n=1}^{N}\sum_{k=1}^{HW} V_{m_{k,n}}}\right)\right), F_m\right) \tag{12}$$

where $\mathcal{C}$ is a concatenation operation, and MLP stands for a multilayer perceptron with two layers. F_1 and F_2 contain each quantization level and the corresponding normalized count. In addition, they contain global average and global maximum features.

To further capture texture-related information more easily, the statistical features are used to improve the representation ability of texture details. The self-attention mechanism is used to reconstruct the statistical features, so that texture information in each new quantization level in the statistical features can be obtained by sensing the statistical information in all the original levels. As a result, the reconstructed statistical features $F_1' \in \mathbb{R}^{C \times H \times W}$ and $F_2' \in \mathbb{R}^{C \times H \times W}$ are reformulated as

$$F_1' = W_1(F_1) \cdot Softmax\left(W_2(F_1)^{\mathrm{T}} \cdot W_3(F_1)\right) \cdot V_a \tag{13}$$

$$F_2' = W_1(F_2) \cdot Softmax\left(W_2(F_2)^{\mathrm{T}} \cdot W_3(F_2)\right) \cdot V_m \tag{14}$$

where W_1, W_2, and W_3 denote three different 1×1 convolutional operations, Softmax is the softmax function, and and V_a and V_m are the quantization encoding matrices corresponding to the global average and maximum, respectively. The statistical texture feature $F_{\text{sta}} \in \mathbb{R}^{C \times H \times W}$ is obtained by fusing the statistical features F_1' and F_2' after texture enhancement. They can be formulated as

$$F_{\text{sta}} = F_1' \oplus F_2' \tag{15}$$

where $\oplus$ is the elementwise adding operation.

By calculating the similarity between each position in the feature map F and the global average feature F_{avg}, as well as the global maximum feature $F_{\max}$, the texture details in low-level features can be more completely displayed, thus making statistical texture features captured after multilevel quantization more discriminative. In addition, a smoother method is used for quantization encoding, which can effectively avoid the problem of gradient vanishing in backpropagation.

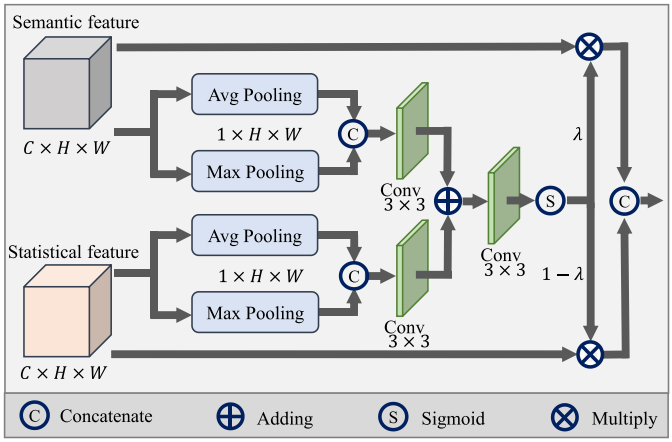


Fig. 4. CFFM's architecture. The semantic features and statistical features are, respectively, fed into average pooling and maximum pooling, and then pass through convolutional layers to obtain the fusion weight λ, which is used to adaptively fuse the two parts of the features.

C. Complete Feature Fusion Module

To better fuse statistical features and semantic features, CFFM is designed to adaptively fuse the two parts of features. Fig. 4 illustrates the CFFM's flowchart.

For the captured semantic features F_{sem} and statistical features F_{sta}, after average pooling and maximum pooling, respectively, the concatenated pooling features are passed through a convolution layer to obtain $F_{\text{sem}}' \in \mathbb{R}^{1 \times H \times W}$ and $F_{\text{sta}}' \in \mathbb{R}^{1 \times H \times W}$. The formulation is expressed as

$$F_{\text{sem}}' = \text{Conv}(\mathcal{C}(A(F_{\text{sem}}), M(F_{\text{sem}}))) \tag{16}$$

$$F_{\text{sta}}' = \text{Conv}(\mathcal{C}(A(F_{\text{sta}}), M(F_{\text{sta}}))) \tag{17}$$

where A and M denote the global average pooling and global max pooling, respectively, $\mathcal{C}$ is a concatenation operation, and Conv here represents the 3×3 convolutional layer. Subsequently, F_{sem}' and F_{sta}' are fed into a convolutional layer after elementwise adding operation, and and weight $\lambda \in \mathbb{R}^{1 \times H \times W}$ is obtained by the Sigmoid activation function. The weight $\lambda \in \mathbb{R}^{1 \times H \times W}$ can be expressed as

$$\lambda = \text{Sigmod}\left(\text{Conv}\left(F_{\text{sem}}' \oplus F_{\text{sta}}'\right)\right) \tag{18}$$

where Sigmod$(\cdot)$ represents the Sigmoid activation function, and $\oplus$ is the elementwise adding operation. After obtaining

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Algorithm 1 HIC by STANet

Input: Hyperspectral image H; training label Y; training epochs Te; multilevel quantization N; and balance factor α.

Output: Classification map P.

```
Randomly initialize all network weights;
while epoch < Te do
    Extract semantic feature F_sem by multiscale semantic feature module with (1)–(4);
    Concatenate the first two layers of features to construct low-features F, and then compute cosine similarity S_a and S_m by (5) and (6);
    Divide the two cosine similarity matrices into N levels Q_a and Q_m by multilevel quantization with (7) and (8);
    Transform the cosine similarity matrices into two quantization encoding matrices V_a and V_m by (9) and (10);
    Obtain statistical features F_1 and F_2 by counting the quantization encoding with (11) and (12);
    Obtain reconstructed statistical features F_1' and F_2' by (13) and (14);
    Obtain the statistical texture feature F_sta by fusing F_1' and F_2' with (15);
    Adaptively fuse F_sem and F_sta to obtain F_spe−spa by (16)–(19);
end while
Train STANet using hybrid loss ℒ with balance factor α;
Predict labels obtain classification map P.
```

the weight λ, the semantic features F_{sem} and statistical features F_{sta} are subsequently fused to obtain the spectral–spatial features $F_{\text{spe-spa}} \in \mathbb{R}^{2C\times H\times W}$ for predicting land cover categories. The spectral–spatial features $F_{\text{spe-spa}} \in \mathbb{R}^{2C\times H\times W}$ can be expressed as

$$F_{\text{spe-spa}} = \mathcal{C}(\lambda \otimes F_{\text{sem}}, (1-\lambda) \otimes F_{\text{sta}}) \tag{19}$$

where $\otimes$ denotes the multiplication operation.

D. Loss Function

To better train STANet, a hybrid loss is used to supervise the end-to-end training. The cross-entropy loss $\mathcal{L}_C$ is commonly used in HIC to update the model parameters. In addition, an online hard examples mining loss $\mathcal{L}_O$ is introduced to alleviate the sample imbalance problem in training. The hybrid loss $\mathcal{L}$ is denoted as

$$\mathcal{L} = \mathcal{L}_C + \alpha \cdot \mathcal{L}_O \tag{20}$$

where α is the balance factor between $\mathcal{L}_C$ and $\mathcal{L}_O$. In $\mathcal{L}_O$, a hard sample refers to the one whose predicted probability corresponding to the correct category is less than θ. At least K pixels are kept for training in each batch to ensure a stable training process. In this article, the balance factor α is set to 0.4. θ and K are set to 0.7 and 100 000, respectively. Algorithm 1 gives the overall training process of STANet.

TABLE I
DATASET DETAILS OF INDIAN PINES (THE BACKGROUND CATEGORY IS NOT INCLUDED)

#	Category	Train	Test	Samples/pixels
1	Alfalfa	3	43	46
2	Corn-notill	72	1356	1428
3	Corn-mintill	42	788	830
4	Corn	12	225	237
5	Grass-pasture	25	458	483
6	Grass-tree	37	693	730
7	Grass-pasture-mowed	2	26	28
8	Hay-windrowed	24	454	478
9	Oats	1	19	20
10	Soybean-notill	49	923	972
11	Soybean-mintill	122	2332	2455
12	Soybean-clean	30	563	593
13	Wheat	11	194	205
14	Woods	64	1201	1265
15	Buildings-Tree-Drives	20	366	386
16	Stone-Steel-Towers	5	88	93
Total		519	9730	10249

III. EXPERIMENTAL STUDY

To verify STANet's effectiveness, we compare it with 11 representative methods on three public datasets. Three quantitative metrics, i.e., overall accuracy (OA), average accuracy (AA), and kappa coefficient (KC), are used to evaluate the performance of all the methods. To ensure impartiality, the experimentation process is repeated five times, and the resulting mean and standard deviation are recorded.

A. Dataset Details

Three publicly available datasets,[1] namely, Indian Pines, Pavia University, and Salinas, are used to evaluate the performance. Tables I–III present the details of such four datasets. The reasons of choosing these three dataset are as follows. First, from the perspective of the dimension of the data, the spatial and spectral dimensions of the three data are different. Second, in terms of land cover categories, these three datasets also contain different land cover categories. Therefore, experiments on these three datasets can meet the diversity of experimental datasets, and more fully demonstrate the effectiveness of the proposed STANet. In addition, the more challenging Houston dataset is selected to further validate the classification performance of the proposed network.

1) Indian Pines: It collects land surface information in Indiana, USA, as captured by the AVRIS sensor. The dimension of the dataset is 145 × 145 × 200, encompassing a total of 16 land cover classes. In the experiment, 5% of samples from each category are randomly selected for training, while the remaining samples are allocated for testing.

2) Pavia University: It encompasses the surface information of the University of Pavia, which is collected by the ROSIS sensors. The dimension of the Pavia University datasets is 610 × 340, accompanied by a total of 103 spectral bands. The Pavia University dataset contains nine categories in total. In the experiment, a subset of 30 samples from each class are

[1] https://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

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TABLE II
DATASET DETAILS OF PAVIA UNIVERSITY (THE BACKGROUND CATEGORY IS NOT INCLUDED)

#	Category	Train	Test	Samples/pixels
1	Asphalt	30	6601	6631
2	Meadows	30	18610	18649
3	Gravel	30	2069	2099
4	Trees	30	3034	3064
5	Painted Metal Sheets	30	1315	1345
6	Bare Soil	30	4999	5029
7	Bitumen	30	1300	1330
8	Self-blocking Brikcs	30	3652	3682
9	Shadows	30	914	947
Total		270	42506	42776

TABLE III
DATASET DETAILS OF SALINAS (THE BACKGROUND CATEGORY IS NOT INCLUDED)

#	Category	Train	Test	Samples/pixels
1	Broccoli-green-weeds-1	30	1979	2009
2	Broccoli-green-weeds-2	30	3696	3726
3	Fallow	30	1946	1976
4	Fallow-rough-plow	30	1364	1394
5	Fallow-smooth	30	2648	2678
6	Stubble	30	3629	3959
7	Celery	30	3549	3579
8	Grapes-untrained	30	11241	11271
9	Soil-vinyard-develop	30	6173	6203
10	Cron-senesced-green-weeds	30	3248	3278
11	Lettuce-romaine-4wk	30	1038	1068
12	Lettuce-romaine-5wk	30	1897	1927
13	Lettuce-romaine-6wk	30	886	916
14	Lettuce-romaine-7wk	30	1040	1070
15	Vinyard-untrained	30	7238	7268
16	Vinyard-vertical-trellis	30	1057	1087
Total		480	53649	54129

randomly extracted to compose the training set, while the rest samples constitute the test set.

3) Salinas: It comprises surface information from Salinas Valley in California, captured by the AVRIS sensor. The SA dataset contains 204 spectral bands with a length and width of 512 × 217. A total of 16 land cover categories are included. Experimental arrangements dictate the random selection of 30 samples from each category to constitute the training set, and the remaining samples are used for testing.

B. Experimental Settings

In the experiments, the stochastic gradient descent (SGD) with momentum is used as the optimizer, where the momentum is set to 0.9 and the weight decay is set to 1e-3. The initial learning rate is 0.01 and the polynomial scheduling policy is used to adjust the learning rate, which is multiplied by $(1 - (\text{iter}/\text{max_iter}))^{0.9}$. Since the whole image is input to the network, the batch size is set to 1 in the training process. The model is trained for a total of 1000 epochs. For the parameter settings in STANet, the small multilevel quantization parameter N means that it is difficult to effectively capture the statistical characteristics of the texture in the shallow features. Larger N may lead to sparse high-dimensional features due to too dense quantization, which may negatively affect the stability of training. Similarly, setting the right balance factor α can help improve the performance and stability of model training. Similarly, choosing the right balance factor α can help improve the performance and stability of model training. After several experimental investigations on the two parameters, the multilevel quantization parameter N is set to 128 and the balance factor α is set to 0.4. The detailed investigation of parameters N and α is presented in Section III-C. Implementation of the proposed STANet architecture is concretized through utilization of a PyTorch framework. Experiment environment is on Ubuntu 18.04.5 and NVIDIA RTX 3090 graphics card.

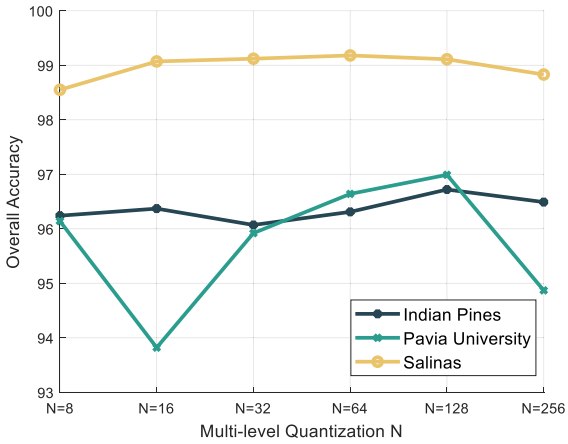


Fig. 5. Experimental results with different values of multilevel quantization parameter N on three datasets.

C. Parameter Investigation

In this section, the value of the multilevel quantization parameter N is investigated. On the three datasets, different N are set for supporting experiments. In the experiments, the values of N are taken from 8 to 256. Fig. 5 presents the OA on the three datasets for different values of N. It can be concluded from Fig. 5 that the OA value is the highest when N is equal to 128 on both the Indian Pines and Pavia University datasets. On the Salinas dataset, the OA value with N being 128 is slightly lower than that with N being 64, fluctuating around 99.00%. Therefore, the multilevel quantization parameter N is set to 128 in all the experiments.

In addition, the values of different balance factors α in the loss function are investigated. On the three datasets, the balance factor α is set to conduct the experiment from 0.1 to 0.5. Fig. 6 presents the classification performance of different balance factor α values on the three datasets. It can be seen from the figure that under different values of balance factor α, the classification accuracy of the model fluctuates. When the value of α is 0.4, the classification performance on the three models is optimal. Therefore, the value of balance factor α in the experiment is 0.4.

D. Comparison With Representative Methods

To verify STANet's effectiveness, the comparative experiments with other nine representative methods are conducted on three datasets. The comparison methods include patch-based and image-based HIC methods.

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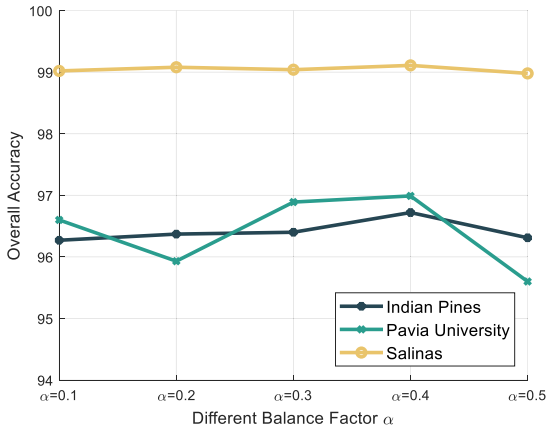


Fig. 6. Experimental results with different values of different balance factors α on three datasets.

TABLE IV
COMPARISON RESULTS WITH PATCH-BASED HIC METHODS ON THE INDIAN PINES DATASET (%)

#	pResNet [34]	SSRN [37]	RSSAN [46]	HybridSN [38]	A2S2KNet [47]	CVSSN [48]	STANet
1	87.12±7.10	5.0±8.67	59.99±9.51	70.45±8.90	81.82±18.80	92.41±9.26	85.11±3.89
2	92.80±1.51	95.12±1.09	92.16±1.96	93.11±1.04	95.34±1.75	93.86±2.78	98.13±0.38
3	95.59±1.78	97.56±0.68	92.41±1.92	94.04±3.57	97.11±1.27	97.46±1.25	96.01±0.73
4	86.96±7.39	88.45±11.73	87.01±4.11	87.45±5.45	88.31±4.21	96.12±3.71	94.04±0.81
5	97.41±1.23	99.17±0.57	93.19±1.92	95.53±1.51	98.58±0.86	93.74±4.26	97.29±0.63
6	98.17±0.38	99.34±0.23	99.33±0.29	97.51±1.41	99.20±0.39	99.09±0.76	96.71±0.93
7	39.91±22.82	0.00±0.00	52.69±12.17	38.88±17.76	67.03±33.15	82.92±16.02	92.30±8.60
8	99.78±0.58	99.98±0.07	100±0.00	99.94±0.11	100±0.00	98.04±3.4	100±0.00
9	83.63±21.13	0.00±0.00	47.78±12.34	35.52±5.04	52.63±38.84	85.78±11.83	68.42±13.41
10	93.02±2.93	93.40±0.99	86.62±2.12	92.06±0.85	91.86±1.33	92.3±3.66	86.98±2.39
11	96.51±1.65	97.87±0.41	95.54±1.17	97.12±0.74	97.46±0.49	94.71±1.48	98.48±0.31
12	91.14±4.57	96.85±1.41	89.17±4.43	87.16±5.42	94.97±2.10	89.39±4.54	96.48±1.12
13	90.77±6.84	89.53±3.41	98.59±2.15	75.25±5.35	91.38±3.40	99.49±0.65	99.07±0.84
14	97.83±0.71	99.65±0.22	97.89±0.68	97.98±1.39	98.98±0.61	96.10±1.6	98.67±0.77
15	90.77±2.80	95.34±1.94	83.93±3.21	88.76±3.16	94.76±2.71	92.11±1.47	98.80±0.79
16	94.44±5.37	93.52±9.70	94.88±5.47	77.56±10.55	98.63±1.29	93.67±3.43	98.18±1.02
OA	94.96±0.74	96.20±0.52	93.31±0.99	93.89±0.55	96.28±0.49	94.77±1.36	**96.72±0.25**
AA	89.74±1.44	78.24±1.45	85.69±1.01	83.02±2.05	90.51±3.94	93.57±1.84	**94.04±1.35**
KC	94.25±0.84	95.67±0.59	92.36±1.14	93.02±0.63	95.75±0.56	94.03±1.55	**96.25±0.29**

1) *Patch-Based HIC Methods:* pResNet [34], SSRN [37], RSSAN [46], HybridSN [38], A2S2KNet [47], and CVSSN [48].
2) *Image-Based HIC Methods:* ENL-FCN [74], SSFCN [71], FContNet-PSP [75], and FContNet-ASPP [75].

1) Results on Indian Pines Dataset: The Indian Pines dataset contains 16 different crops, and there is large distribution imbalance among classes, which is a challenging dataset. The experimental results of this dataset are shown in Tables IV and V. It can be seen from Table IV and V that A2S2KNet achieves a higher OA value among the patch-based methods. In the image-based methods, the OA value of ENL-FCN is slightly lower than that of A2S2KNet. The OA value of the proposed STANet is 96.59%, which is 0.31% and 0.58% higher than those of A2S2KNet and ENL-FCN, respectively. Although the AA value of STANet is lower than CVSSN's, which may be due to the fact that the ninth class only contains 20 samples, making it difficult for the model to fully capture the discriminative features, thus resulting in a low accuracy on the ninth class. However, STANet achieves high accuracy on the other categories. The KC value of STANet on the Indian Pines dataset also exceeds that of the comparative methods.

To further demonstrate the effectiveness of STANet, the classification results are visualized in Fig. 7. The ground truth of the Indian Pines dataset and the visual classification map

TABLE V
COMPARISON RESULTS WITH IMAGE-BASED HIC METHODS ON THE INDIAN PINES DATASET (%)

#	ENL-FCN [74]	SSFCN [71]	FContNet-PSP [75]	FContNet-ASPP [75]	STANet
1	94.88±5.04	47.44±16.32	88.37±11.03	81.86±3.82	85.11±3.89
2	91.50±1.73	57.66±4.67	92.97±0.53	94.26±2.50	98.13±0.38
3	96.29±2.50	55.83±7.48	95.07±2.81	94.74±4.19	96.01±0.73
4	84.35±8.71	57.42±5.73	81.86±9.89	93.33±4.34	94.04±0.81
5	98.34±0.86	60.48±8.44	92.7±6.31	89.73±7.65	97.29±0.63
6	96.13±1.78	78.61±4.82	97.40±1.45	98.93±1.29	96.71±0.93
7	93.07±7.88	65.38±18.24	70.76±41.56	92.30±10.87	92.30±8.60
8	99.51±0.73	94.01±5.16	99.95±0.09	99.16±1.40	100±0.00
9	0.00±0.00	42.11±23.83	63.15±38.49	57.89±18.60	68.42±13.41
10	96.09±1.78	65.98±5.07	93.84±4.36	91.67±2.64	86.98±2.39
11	97.61±0.72	74.76±8.20	97.77±0.42	98.00±1.88	98.48±0.31
12	96.30±1.83	48.52±7.46	85.54±10.44	90.51±3.35	96.48±1.12
13	97.31±2.04	88.86±4.76	98.14±1.65	96.18±3.45	99.07±0.84
14	98.65±0.38	86.67±5.87	99.86±0.17	99.41±0.38	98.67±0.77
15	98.03±0.52	50.49±5.75	89.34±8.36	98.25±1.47	98.80±0.79
16	91.81±2.32	82.5±15.77	89.54±9.38	84.31±21.25	98.18±1.02
OA	96.01±1.11	69.29±2.74	94.96±1.04	95.66±0.43	**96.72±0.25**
AA	89.37±1.52	66.04±1.69	89.77±3.77	91.28±1.91	**94.04±1.35**
KC	95.46±1.26	65.05±2.93	94.26±1.19	95.05±0.49	**96.25±0.29**

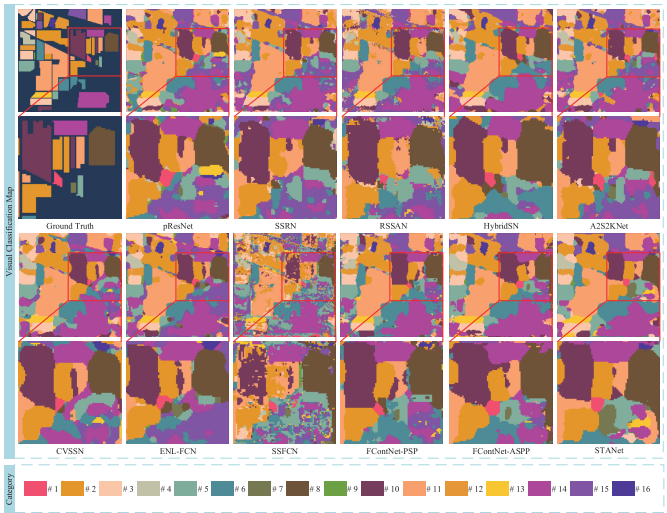


Fig. 7. Visual classification map of different methods on the Indian Pines dataset.

of comparison methods are presented. From Fig. 7, it can be concluded that among the patch-based comparison methods, the visualization results of SSRN and A2S2KNet are closer to ground truth. There are many misclassifications in the visual classification map of SSFCN. In contrast, the visualization result of STANet, which is more pure, has less difference from ground truth and has fewer misclassification pixels.

2) Results on the Pavia University Dataset: The Pavia University dataset contains nine categories of urban ground objects, and the distribution of most categories has an irregular shape. Comparison results with the patch-based and image-based HIC methods are presented in Tables VI and VII. It can be concluded that A2S2KNet achieves an OA value of 94.83%. In the image-based methods, the OA value of FContNet-ASPP is 1.00% higher than that of A2S2KNet. STANet outperforms FContNet-ASPP and A2S2KNet in all three evaluation metrics on the Pavia University dataset. The OA value is 1.11% and 2.11% higher, which indicates that STANet can correctly predict the labels of most examples. The AA value exceeds 0.88% and 0.82%, respectively. This shows that STANet can obtain higher accuracy on most categories, especially in the fifth, seventh, and ninth categories, which has obtained more

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TABLE VI

COMPARISON RESULTS WITH PATCH-BASED HIC METHODS ON THE PAVIA UNIVERSITY DATASET (%)

#	pResNet [34]	SSRN [37]	RSSAN [46]	HybridSN [38]	A2S2KNet [47]	CVSSN [48]	STANet
1	93.48±3.01	94.55±1.63	83.18±3.21	85.18±5.39	97.24±1.56	97.89±1.16	94.52±0.62
2	95.39±1.70	94.21±2.28	96.56±2.46	96.12±4.75	93.44±4.32	96.90±0.42	98.22±0.56
3	87.46±3.07	86.49±5.77	84.68±5.58	89.97±6.48	90.21±3.48	83.40±6.32	97.11±1.49
4	94.78±2.32	94.78±2.27	93.04±4.16	91.31±3.24	98.12±0.94	91.61±4.50	94.67±1.68
5	99.81±0.12	99.97±0.06	99.88±0.12	99.34±0.96	99.92±0.16	96.18±3.33	99.82±0.09
6	96.63±2.60	96.20±3.28	97.45±0.66	98.77±1.90	94.30±2.95	62.80±7.10	96.58±1.53
7	98.01±2.03	98.97±1.32	99.24±0.48	99.64±0.40	98.63±1.18	82.99±9.54	99.71±0.34
8	89.75±2.53	91.66±1.89	81.37±3.40	87.80±5.80	93.66±1.77	88.76±7.66	95.12±2.41
9	98.73±0.78	99.89±0.13	97.22±2.82	97.03±1.48	99.91±0.14	96.56±1.93	99.04±0.80
OA	94.61±0.23	94.39±0.57	92.65±1.63	93.61±1.44	94.83±1.88	88.51±2.58	**96.99±0.27**
AA	94.90±0.37	95.19±0.91	92.51±1.45	93.91±0.84	96.16±0.81	88.57±1.44	**97.20±0.55**
KC	92.90±0.29	92.62±0.72	90.31±2.12	91.58±1.80	93.22±2.41	85.17±3.16	**96.01±0.36**

TABLE VII

COMPARISON RESULTS WITH IMAGE-BASED HIC METHODS ON THE PAVIA UNIVERSITY DATASET (%)

#	ENL-FCN [74]	SSFCN [71]	FContNet-PSP [75]	FContNet-ASPP [75]	STANet
1	91.78±2.77	79.26±6.28	91.43±3.77	89.64±1.69	94.52±0.62
2	93.86±2.01	88.67±2.85	91.87±6.83	97.83±1.24	98.22±0.56
3	92.89±1.86	79.10±4.83	90.87±5.72	94.92±3.33	97.11±1.49
4	94.95±0.92	94.71±2.17	97.22±1.40	93.90±2.58	94.67±1.68
5	99.89±0.08	96.63±2.21	99.57±0.65	99.93±0.09	99.82±0.09
6	89.29±3.78	90.21±5.66	96.15±2.42	97.68±3.93	96.58±1.53
7	98.03±0.59	92.38±4.34	99.35±0.12	98.52±1.54	99.71±0.34
8	95.04±1.64	80.66±8.83	95.19±2.36	93.13±5.04	95.12±2.41
9	99.67±0.20	99.80±0.27	99.76±0.24	99.38±0.65	99.04±0.80
OA	93.57±0.89	87.25±2.61	93.56±2.85	95.83±0.49	**96.99±0.27**
AA	95.04±0.58	89.05±2.30	95.71±0.56	96.10±0.69	**97.20±0.55**
KC	91.53±1.16	83.43±3.37	91.62±3.58	94.48±0.65	**96.01±0.36**

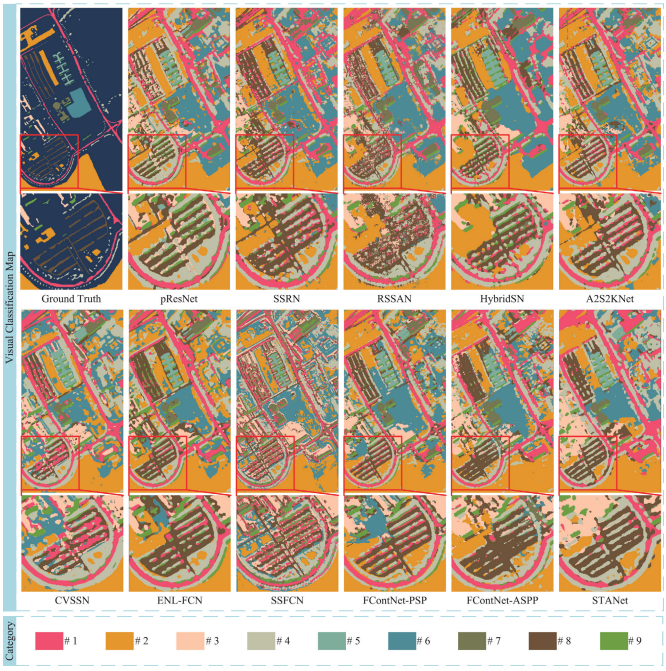


Fig. 8. Visual classification map of different methods on the Pavia University dataset.

than 99.00% of the classification accuracy. The KC value is over 1.20 and 2.46, respectively, which indicates that the classification results of STANet have higher consistency with ground truth.

Fig. 8 illustrates the ground truth and visual classification maps of other methods for the Pavia University dataset. It can be seen from Fig. 8 that in the patch-based comparison methods, there are more misclassified pixels in the classification map of CVSSN. In the image-based methods, the visualization results of SSFCN are more different from ground truth. In contrast, the visualization results of STANet on the Pavia

TABLE VIII

COMPARISON RESULTS WITH PATCH-BASED HIC METHODS ON THE SALINAS DATASET (%)

#	pResNet [34]	SSRN [37]	RSSAN [46]	HybridSN [38]	A2S2KNet [47]	CVSSN [48]	STANet
1	99.84±0.10	100±0.00	99.99±0.02	100±0.00	99.98±0.04	99.34±0.66	96.83±1.68
2	99.96±0.05	99.96±.0.06	99.94±0.06	100±0.00	99.98±0.03	99.97±0.07	99.98±0.02
3	99.85±0.20	100±0.00	99.87±0.20	100±0.00	100±0.00	96.62±1.63	100±0.00
4	99.44±0.80	99.85±0.13	97.82±1.24	99.67±0.42	99.92±0.12	98.49±0.85	98.44±0.62
5	99.23±0.79	98.43±0.79	95.10±1.69	99.51±0.40	99.26±0.27	98.21±1.26	98.85±0.13
6	99.51±1.09	100±0.00	99.95±0.04	99.28±1.39	99.87±0.27	99.76±0.32	99.89±0.15
7	99.94±0.04	100±0.00	99.06±0.94	99.83±0.20	100±0.00	99.51±0.30	99.96±0.03
8	84.58±2.93	88.53±2.57	89.50±4.30	90.56±0.98	90.12±2.59	90.66±3.75	99.25±0.20
9	99.99±0.01	99.98±0.02	99.44±0.35	99.93±0.08	100±0.00	99.56±0.45	99.98±0.03
10	95.99±1.71	96.01±1.52	95.98±1.82	99.53±0.68	97.21±1.95	91.33±7.28	99.92±0.11
11	99.76±0.34	99.57±0.41	99.63±0.54	99.63±0.44	99.82±0.38	89.45±5.53	99.85±0.15
12	99.94±0.09	99.50±1.02	99.87±0.16	99.94±0.06	99.99±0.02	99.42±0.91	99.93±0.17
13	98.37±2.78	100±0.00	98.12±2.68	98.80±1.83	100±0.00	98.56±0.95	99.86±0.30
14	99.61±0.19	99.72±0.28	97.26±3.00	99.46±1.15	99.80±0.27	98.01±1.09	98.21±0.53
15	89.89±2.34	82.05±2.76	92.57±3.42	94.92±3.84	90.78±1.94	76.33±5.40	96.02±0.56
16	98.24±0.70	98.37±0.44	98.87±0.66	99.74±0.57	98.84±0.57	97.38±1.50	100 ±0.00
OA	94.95±0.66	94.76±0.35	95.98±0.56	97.15±0.50	96.42±0.52	93.06±0.79	**99.11±0.06**
AA	97.76±0.42	97.62±0.11	97.68±0.28	98.80±0.15	98.47±0.20	95.79±0.61	**99.26±0.11**
KC	94.39±0.73	94.17±0.38	95.53±0.62	96.83±0.55	96.01±0.58	92.29±0.86	**99.01±0.06**

TABLE IX

COMPARISON RESULTS WITH IMAGE-BASED HIC METHODS ON THE SALINAS DATASET (%)

#	ENL-FCN [74]	SSFCN [71]	FContNet-PSP [75]	FContNet-ASPP [75]	STANet
1	100±0.00	97.55±1.78	100±0.00	99.05±1.03	96.83±1.68
2	99.92±0.14	94.40±2.49	99.70±0.46	98.82±1.12	99.98±0.02
3	99.98±0.02	98.40±0.75	100±0.00	99.66±0.46	100±0.00
4	99.36±0.39	98.62±2.10	98.75±0.92	99.89±0.16	98.44±0.62
5	99.44±0.08	93.36±3.84	98.41±1.00	98.30±1.77	98.85±0.13
6	100±0.00	97.03±1.71	99.93±0.13	99.57±0.76	99.89±0.15
7	99.65±0.54	93.15±2.34	99.89±0.14	99.90±0.09	99.96±0.03
8	86.64±2.07	74.30±9.93	91.16±5.98	92.83±1.92	99.25±0.20
9	99.87±0.17	95.32±4.15	99.91±0.08	99.93±0.10	99.98±0.03
10	97.37±0.65	85.79±2.54	98.90±0.37	97.41±1.77	99.92±0.11
11	100±0.00	96.33±1.48	99.49±0.69	99.94±0.08	99.85±0.15
12	100±0.00	97.12±1.70	99.31±0.66	99.12±1.19	99.93±0.17
13	100±0.00	95.93±1.10	99.66±0.44	100±0.00	99.86±0.30
14	99.76±0.14	95.23±2.01	99.65±0.29	99.78±0.25	98.21±0.53
15	92.29±3.10	71.97±10.85	95.53±4.33	96.54±0.85	96.02±0.56
16	98.19±0.23	86.16±10.92	98.07±3.26	100±0.00	100 ±0.00
OA	95.76±0.79	87.08±3.27	97.21±0.92	97.49±0.50	**99.11±0.06**
AA	98.25±0.36	91.92±2.05	98.65±0.27	98.80±0.24	**99.26±0.11**
KC	95.28±0.88	85.65±3.61	96.90±1.02	97.30±0.50	**99.01±0.06**

University dataset reveal fewer misclassified pixels and higher accuracy.

3) Results on Salinas Dataset: The dataset contains crops such as broccoli, fallow, and celery. From Table VIII, the OA value of FContNet-ASPP is the highest in the comparison method and the OA value of HybridSN is 97.15%. The OA value of STANet is 98.90%, which exceeds HybridSN and FContNet-ASPP. The AA value of STANet also exceeds 0.16%. As seen from Table IX, STANet can effectively predict the class label in most categories. It can also be seen from KC that the classification results of STANet have higher consistency with ground truth than other comparison methods.

The ground truth of the Salinas dataset and the visual classification maps of different methods are presented in Fig. 9. From the visual classification maps, it can be seen that HybridSN and FContNet-ASPP have the least misclassified pixels. Compared with the proposed STANet, we see that the visualization results of the proposed method are closer to the ground truth, with fewer misclassified pixels and a cleaner classification map, as shown in Fig. 9.

E. Ablation Study

To verify the effectiveness of the proposed module, several ablation experiments are conducted on three datasets. STLM is constructed to extract the statistical texture information of the low-level features. CFFM is designed to adaptively

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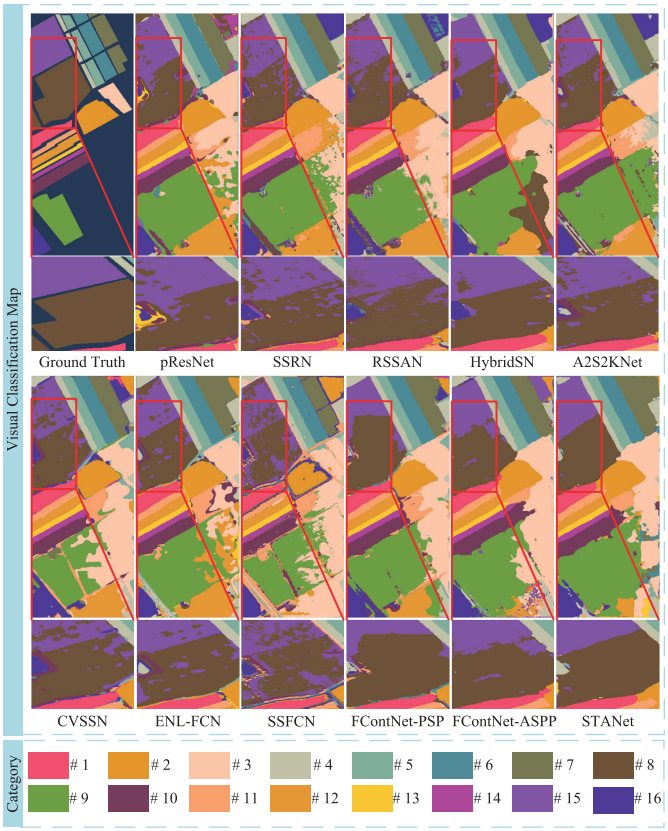


Fig. 9. Visual classification map of different methods on the Salinas dataset.

TABLE X
RESULTS OF ABLATION EXPERIMENTS ON THREE DATASETS (%)

Baseline	STLM	CFFM	Iindian Pines	Pavia University	Salinas
✓			96.12	94.81	98.46
✓	✓		96.49	96.38	98.54
✓	✓	✓	96.72	96.99	99.11

fuse semantic and statistical features. The results of ablation experiments on the three datasets are reported in Table X. Baseline refers to the network architecture that does not include STLM and CFFM in the model. Obviously, the OA value of the baseline on the three datasets is 96.12%, 94.81%, and 98.46%, respectively. This demonstrates that the baseline can effectively capture the spectral–spatial feature of hyperspectral images. After the introduction of STLM, the OA value on the three datasets is increased by 0.37%, 1.57%, and 0.12%, respectively. It can be seen that the discrimination of the spectral–spatial features is improved after introducing the low-level features from statistical texture information.

The comprehensiveness of feature modeling is enhanced from a feature-level perspective, encompassing both high-level semantic and low-level statistical texture features. To optimize the fusion of statistical texture and semantic features, CFFM is devised to learn an adaptive weight, denoted as λ, for feature fusion. In comparison to concatenating the two-part features, wherein the weight λ equals 0.5, and the two features are fused, the embedding of CFFM results in an improvement in classification performance by 0.23% on the Indian Pines dataset, 0.61% on the Pavia University dataset, and 0.47% on the Salinas dataset. This substantiates the effectiveness of the designed CFFM for adaptive feature fusion.

TABLE XI
RESULTS OF ABLATION EXPERIMENTS ON DIFFERENT SIMILARITY MEASURES (%)

	Indian Pines	Pavia University	Salinas
average	95.87	95.74	98.70
maximum	96.39	95.60	98.79
average+maximumax	96.72	96.99	99.11

TABLE XII
RESULTS OF ABLATION EXPERIMENTS ON DIFFERENT FEATURE AGGREGATION METHODS (%)

	Indian Pines	Pavia University	Salinas
Adding	96.27	95.92	98.51
Concatenation	96.72	96.99	99.11

The calculation of similarity between average and maximum values is undertaken when quantifying the texture information of shallow features at multiple levels, facilitating the capture of more extensive statistical texture features. To validate the efficacy of amalgamating mean and maximum values, ablation experiments using only average or maximum values are conducted on three datasets individually. Table XI presents the outcomes of this ablation experiment. It is evident from the table that the result score achieved by combining the mean and the maximum values significantly outperforms the capture of a single statistical texture feature. This underscores that the amalgamation of statistical texture features, specifically mean and maximum values, enables the acquisition of more comprehensive texture information, thereby enhancing the classification performance of the network.

In the process of feature fusion, the concatenation operation is used to merge features. Ablation experiments are conducted on two feature fusion strategies across three datasets, namely, the additive operation and the concatenation operation. Table XII illustrates the experimental outcomes under these two strategies. It is conducted from the table that the performance achieved with the concatenation strategy surpasses that of the additive and summation operations by 0.45%, 1.07%, and 0.6%, respectively. Consequently, the concatenation operation is adopted for feature fusion.

In addition, the effectiveness of introducing OHEM loss is verified. The performance of using loss functions $\mathcal{L}_{CE}$ and $\mathcal{L}_{\text{CE+OHEM}}$ to supervise model training is compared on three datasets. Fig. 10 illustrates the experimental results under the supervision of two loss functions. It can be seen that the classification performance of STANet with supervised training of $\mathcal{L}_{\text{CE+OHEM}}$ improves with respect to the supervision of $\mathcal{L}_{CE}$. On three datasets, the OA is increased by 0.34%, 0.42%, and 1.11%, respectively. It is demonstrated that the introduction of OHEM loss can help STANet to learn samples that are difficult to classify and alleviate the problem of sample imbalance, thereby boosting the feature extraction ability of STANet.

To further verify the effectiveness of the statistical properties of texture information, the ablation experiments are conducted on the statistical and structural properties of texture information. In STANet, the proposed STLM module captures statistical properties in low-level features that contain rich

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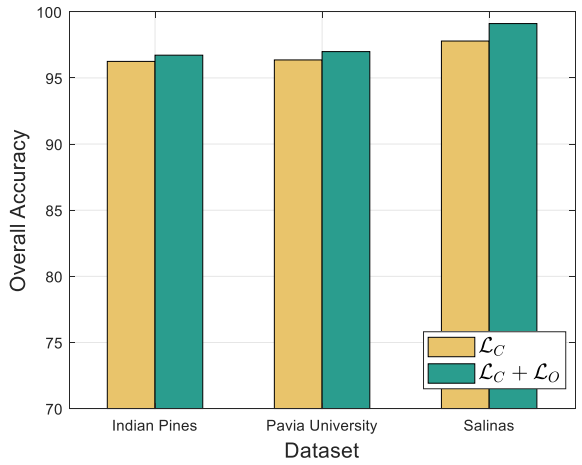


Fig. 10. Experimental results of loss functions $\mathcal{L}_C$ and $\mathcal{L}_C + \mathcal{L}_O$ on three datasets.

TABLE XIII
RESULTS OF ABLATION EXPERIMENTS ON STATISTICAL PROPERTY (%)

	Indian Pines	Pavia University	Salinas
Structural property	96.45	95.24	98.73
Statistical property	96.72	96.99	99.11

texture information. The skip connection is used to introduce the structural properties of low-level features into multiscale semantic features. Table XIII reports the ablation experimental results on the statistical property. It can be concluded from that compared with the introduction of the statistical property, the classification accuracy of three datasets is improved by 0.27%, 1.75%, and 0.38%, respectively, by introducing the statistical property of texture information by STLM. Compared with structural property, statistical property can describe the distribution of different texture details in more detail, thus providing more abundant information to improve the discrimination of features.

F. Training Set Size

In this section, the classification performance under different numbers of training samples is explored. For the Indian Pines dataset, the training sample rate spans from 1% to 10%, while for the Pavia University and Salinas datasets, the training sample rate escalates from 10 to 100. Fig. 11 illustrates the classification results of various HIC methods with fluctuations in the number of training samples across the three datasets. The enlarged display in the partial red frame on the right provides a closer examination. As discernible from Fig. 11, when the proportion of training samples is limited, the proposed network exhibits a significant improvement compared with other HIC methods. For instance, on the Indian Pines dataset, the OA value of the proposed network approaches 94.00%, while the OA value of other comparison methods, such as FContNet-ASPP, only approximates 92.00%. The notable enhancement in the classification performance gap is evident. Although the performance improvement becomes less pronounced as the training sample rate increases, the proposed network remains competitive with other representative counterparts. In general,

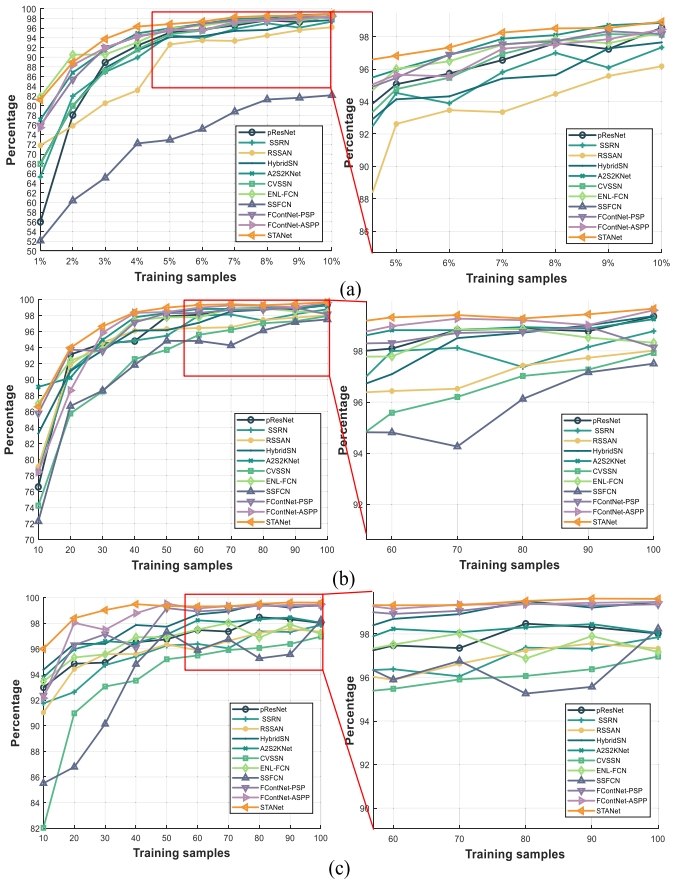


Fig. 11. Experimental results with different training set sizes on three datasets. (a) Indian Pines. (b) Pavia University. (c) Salinas.

TABLE XIV
COMPARISON RESULTS WITH REPRESENTATIVE HIC METHODS ON THE HOUSTON DATASET (%)

	OA	AA	KC
pResNet [34]	81.78±0.86	82.66±1.42	80.29±0.94
SSRN [37]	86.83±2.01	88.71±1.89	85.76±2.16
RSSAN [46]	83.81±1.51	85.81±1.48	82.51±1.62
HybridSN [38]	87.93±0.47	89.47±0.32	86.95±0.51
A2S2KNet [47]	87.56±2.03	89.63±1.61	86.56±2.17
CVSSN [48]	91.68±0.90	91.48±1.05	91.04±0.98
ENL-FCN [74]	92.19±0.23	93.41±0.25	91.56±0.25
SSFCN [71]	78.85±1.94	79.87±2.10	77.15±2.09
FContNet-PSP [75]	92.94±0.49	93.83±0.45	92.36±0.53
FContNet-ASPP [75]	92.01±0.94	93.01±0.74	91.36±1.02
STANet	**93.46±0.45**	**94.59±0.34**	**92.94±0.49**

the classification performance of the proposed network demonstrates significant improvement compared with its peers in the overall trend.

G. Comparison Experiments on the Houston Dataset

To further verify the effectiveness of STANet, comparative experiments are conducted on the more challenging Houston dataset. This dataset contains a total of 144 spectral bands with a spatial dimension of 349 × 1905. It contains 15 categories. In the experiments, 30 samples from each category are randomly selected for training, and the remaining samples are used for testing. The experimental results on the Houston dataset are reported in Table XIV. It can be concluded from Table XIV that CVSSN has the highest classification

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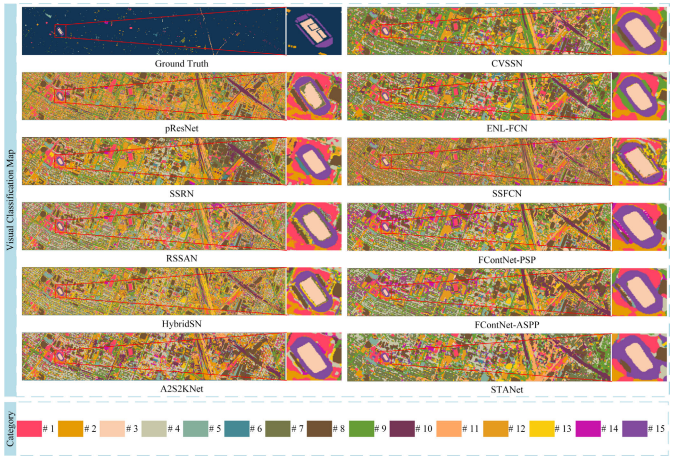


Fig. 12. Visual classification map of different methods on the Houston dataset.

TABLE XV
COMPARISON RESULTS WITH OTHER REPRESENTATIVE-AWARE NETWORK-BASED HIC METHODS (%)

		MSC-EDKELM [55]	NSCKL [61]	GAHT [54]	STANet
Indian Pines	OA	96.19±0.39	95.83±0.77	95.07±0.49	**96.72±0.25**
	AA	94.39±3.23	91.50±1.12	87.02±0.98	**94.04±1.35**
	KC	95.66±0.45	95.24±0.89	94.38±0.56	**96.25±0.29**
Pavia University	OA	96.62±0.37	92.58±0.79	88.66±1.43	**96.99±0.27**
	AA	97.30±0.12	91.75±0.56	83.07±1.21	**97.20±0.55**
	KC	96.42±0.31	90.23±1.01	84.92±1.86	**96.01±0.36**
Salinas	OA	95.65±0.69	98.51±0.80	92.59±1.12	**99.11±0.06**
	AA	97.48±0.36	99.03±0.34	94.55±0.67	**99.26±0.11**
	KC	95.16±0.77	98.34±0.89	91.76±1.24	**99.01±0.06**
Houston	OA	/	92.94±0.96	88.34±1.16	**93.46±0.45**
	AA	/	93.91±0.78	89.24±1.19	**94.59±0.34**
	KC	/	92.37±1.04	87.39±1.25	**92.94±0.49**

accuracy among the comparison methods based on patches, but STANet is better than CVSSN by 1.78%, 3.11%, and 1.90% in three evaluation indicators, respectively. Compared with the FContNet-PSP based on the whole image, the OA value of STANet is 0.52% higher, the AA value is 0.76% higher, and the KC value is 0.58% higher. Fig. 12 shows the visual classification map on the Houston dataset, where the red box is zoomed on the right. It can be intuitively seen from Fig. 12 that there is less noise in the visual map of STANet and fewer pixels are misclassified compared with other methods.

H. Compared With Other Representative-Aware Network-Based HIC Methods

In this section, the comparison experiments with other representative-aware network-based HIC methods are carried out. These methods refer to MS-EDKELM [55], NSCKL [61], and GAHT [54]. Table XV reports the comparative results of STANet with representative-aware network-based HIC methods on four datasets. It can be drawn from Table XV that for the Indian Pines dataset, STANet outperforms other HIC methods in terms of OA values, especially exceeding MSC-EDKELM by 0.53%. For the Pavia University dataset, the OA value of STANet exceeds MSC-EDKELM by 0.37%. For the Salinas dataset, NSCKL has the best performance among the comparative methods, and STANet outperforms it by 0.6%. For the more challenging Houston dataset, STANet outperforms NSCKL and GAHT in terms of three evaluation metrics. When we perform MS-EDKELM methods on the Houston dataset, it is a great pity that due to the hardware limitations of the computing platform, there is a shortage of memory, so we could not obtain the experimental results of MSC-EDKELM on the Houston dataset.

IV. CONCLUSION

In this article, a novel STANet is proposed for HIC. Diverging from the existing methodologies that incorporate shallow texture features into deeper layers through skip connections, this approach establishes a trainable pathway, seamlessly integrating the statistical properties of texture information into the model's parameter learning process. To achieve this, the STLM is proposed, which models the statistical properties of texture information through multilevel quantization and quantization encoding of low-level features. In addition, the CFFM is designed to adaptively merge semantic and statistical features of hyperspectral images. This feature-level perspective enables STANet to simultaneously capture semantic and statistical features, thereby enhancing feature extraction completeness. The experimental results on three public datasets demonstrate that the proposed STANet outperforms other peers in terms of classification performance.

In the further work, starting from time–frequency transformation, we will explore the texture feature representation of hyperspectral images in learnable frequency domains. Thus, we realize the joint learning representation of both the spatial and frequency domains of hyperspectral images. So that it can achieve more accurate ground object category recognition in practical tasks such as precision agriculture and urban planning.

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